Linear Algebra I – Week 6

Tashkent University of Technology

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Sequence and references

20 original worked problems, increasing from 2×2 to 3×3 and advanced factorizations. Problem statements are centered on separate slides. Based on the methods in Strang, *Introduction to Linear Algebra*, 4th ed., Sections 2.4–2.7.

Problem 1 – 2 by 2: inverse

Find A^{-1} for $A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$. Verify by multiplication.

Solution 1 – 2 by 2: inverse

$\det A = 4 \neq 0$. By the 2×2 formula, $A^{-1} = \begin{bmatrix} \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$. Check $AA^{-1} = I$.

Problem 2 – 2 by 2: solve

Solve $Ax = b$ with $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $b = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$.

Solution 2 – 2 by 2: solve

Using elimination or A^{-1}, $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Check $Ax = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$.

Problem 3 – 2 by 2: inverse

Find A^{-1} for $A = \begin{bmatrix} 4 & 1 \\ 2 & 5 \end{bmatrix}$. Verify by multiplication.

Solution 3 – 2 by 2: inverse

$\det A = 18 \neq 0$. By the 2×2 formula, $A^{-1} = \begin{bmatrix} \frac{5}{18} & -\frac{1}{18} \\ -\frac{1}{9} & \frac{2}{9} \end{bmatrix}$. Check $AA^{-1} = I$.

Problem 4 – 2 by 2: solve

Solve $Ax = b$ with $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}$ and $b = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$.

Solution 4 – 2 by 2: solve

Using elimination or A^{-1}, $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Check $Ax = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$.

Problem 5 – 2 by 2: factorization

Find $A = LU = LDU$ for $A = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$.

Solution 5 – 2 by 2: factorization

$L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. Extract pivots: $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ and normalized upper $\widehat{U} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$.

Problem 6 – 2 by 2: factorization

Find $A = LU = LDU$ for $A = \begin{bmatrix} 3 & 1 \\ 3 & 5 \end{bmatrix}$.

Solution 6 – 2 by 2: factorization

$L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$. Extract pivots: $D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ and normalized upper $\widehat{U} = \begin{bmatrix} 1 & \frac{1}{3} \\ 0 & 1 \end{bmatrix}$.

Problem 7 – 2 by 2: factorization

Find $A = LU = LDU$ for $A = \begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$.

Solution 7 – 2 by 2: factorization

$L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$. Extract pivots: $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and normalized upper $\widehat{U} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Problem 8 – 2 by 2: factorization

Find $A = LU = LDU$ for $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$.

Solution 8 – 2 by 2: factorization

$L = \begin{bmatrix} 1 & 0 \\ \frac{5}{2} & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 \\ 0 & \frac{3}{2} \end{bmatrix}$. Extract pivots: $D = \begin{bmatrix} 2 & 0 \\ 0 & \frac{3}{2} \end{bmatrix}$ and normalized upper $\widehat{U} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$.

Problem 9 – 3 by 3: elimination

Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$.

Solution 9 – 3 by 3: elimination

Eliminate entries below the pivots and back-substitute. The solution is $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verify $Ax = b$ directly.

Problem 10 – 3 by 3: elimination

Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$.

Solution 10 – 3 by 3: elimination

Eliminate entries below the pivots and back-substitute. The solution is $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verify $Ax = b$ directly.

Problem 11 – 3 by 3: elimination

Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}$.

Solution 11 – 3 by 3: elimination

Eliminate entries below the pivots and back-substitute. The solution is $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verify $Ax = b$ directly.

Problem 12 – 3 by 3: elimination

Solve $Ax = b$ using Gaussian elimination for $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$.

Solution 12 – 3 by 3: elimination

Eliminate entries below the pivots and back-substitute. The solution is $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verify $Ax = b$ directly.

Problem 13 – 3 by 3: LU and LDU

Find LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$.

Solution 13 – 3 by 3: LU and LDU

Elimination gives $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. The diagonal factor is $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; normalized upper factor is $D^{-1}U = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$.

Problem 14 – 3 by 3: LU and LDU

Find LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$.

Solution 14 – 3 by 3: LU and LDU

Elimination gives $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$. The diagonal factor is $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$; normalized upper factor is $D^{-1}U = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix}$.

Problem 15 – 3 by 3: LU and LDU

Find LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$.

Solution 15 – 3 by 3: LU and LDU

Elimination gives $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. The diagonal factor is $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$; normalized upper factor is $D^{-1}U = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{4} \\ 0 & 0 & 1 \end{bmatrix}$.

Problem 16 – 3 by 3: LU and LDU

Find LU and LDU for $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$.

Solution 16 – 3 by 3: LU and LDU

Elimination gives $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. The diagonal factor is $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; normalized upper factor is $D^{-1}U = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$.

Problem 17 – Row exchange and PLU

Find a row permutation P and factors satisfying $PA = LU$ for

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}.$$

Solution 17 – Row exchange and PLU

Choose $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then $L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & \frac{1}{2} & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{3}{2} \end{bmatrix}$, giving $PA = LU$.

Problem 18 – Transpose and inverse

For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$, compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

Solution 18 – Transpose and inverse

$$A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}; (A^T)^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{2}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} & \frac{4}{5} \\ \frac{2}{5} & -\frac{1}{5} & \frac{1}{5} \end{bmatrix}$$, which equals $(A^{-1})^T$.

Problem 19 – Symmetric LDL transpose

Factor symmetric $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ as LDL^T.

Solution 19 – Symmetric LDL transpose

$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & \frac{1}{3} & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{8}{3} \end{bmatrix}$. Verify $LDL^T = A$.

Problem 20 – Challenge: tridiagonal model

Solve $Ax = b$ by LU for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest-neighbor structure.

Solution 20 – Challenge: tridiagonal model

Factor $A = LU$, solve $Ly = b$, then $Ux = y$. Result $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Only nearest-neighbor entries are nonzero.